Agile Development Process and Devops Lab

FAT EXAMINATION

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Smart Electricity Billing System

Problem Statement :

Ensure billing usage is calculated and billed correctly.

Tasks:

1. Develop billing calculation logic

2. Write Junit tests for tariff calculations

3.Implement CI/CD pipeline with automated testing

4.Deploy services on Docker and Kubernetes

AIM

To implement a CI/CD pipeline using Jenkins, Docker, and Kubernetes with JUnit testing integration for a Java-based Prescription Management System.

1.Java Application

App.java

```java
public class App {

    private static final double BASE_CHARGE = 100.0;
    private static final double RATE_PER_UNIT = 7.0;

    public double calculateBill(double units, boolean isPeakHour, int
tariffLevel) {

        double tariffMultiplier = getTariffMultiplier(isPeakHour, tariffLevel);

        return BASE_CHARGE + (units * RATE_PER_UNIT * tariffMultiplier);
    }

    public double getTariffMultiplier(boolean isPeakHour, int tariffLevel) {

        double tariff = 1.0;

        if (isPeakHour) {
            tariff += 0.5;
        }

        if (tariffLevel > 5) {
            tariff += 0.5;
        }

        return tariff;
    }
}
```

2.JUnit Test Cases

AppTest.java

```
import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;

import org.junit.jupiter.api.BeforeEach;



public class AppTest {



    private App service;



    @BeforeEach

    public void setUp() {

        service = new App();

    }



    @Test

    public void testNormalTariff() {

        double bill = service.calculateBill(10, false, 3);

        assertEquals(170.0, bill, 0.01);

    }



    @Test

    public void testPeakHourTariff() {
```

```
        double bill = service.calculateBill(10, true, 3);
        assertEquals(205.0, bill, 0.01);
    }

    @Test
    public void testHighTariff() {
        double bill = service.calculateBill(10, false, 7);
        assertEquals(205.0, bill, 0.01);
    }

    @Test
    public void testPeakAndHighTariff() {
        double bill = service.calculateBill(10, true, 8);
        assertEquals(240.0, bill, 0.01);
    }

    @Test
    public void testZeroUnits() {
        double bill = service.calculateBill(0, true, 8);
        assertEquals(100.0, bill, 0.01);
    }
}
```

3. Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">

 <modelVersion>4.0.0</modelVersion>

 <groupId>com.demo</groupId>
 <artifactId>test-java-app</artifactId>
 <version>1.0</version>

 <properties>
  <maven.compiler.source>17</maven.compiler.source>
  <maven.compiler.target>17</maven.compiler.target>
  <junit.version>5.10.1</junit.version>
 </properties>

 <dependencies>
  <dependency>
   <groupId>org.junit.jupiter</groupId>
   <artifactId>junit-jupiter</artifactId>
   <version>${junit.version}</version>
   <scope>test</scope>
  </dependency>
 </dependencies>

 <build>
  <plugins>

   <!-- Plugin to run JUnit tests -->
   <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-plugin</artifactId>
    <version>3.2.5</version>
   </plugin>

   <!-- Plugin to create executable JAR -->
```

```
  <plugin>
   <groupId>org.apache.maven.plugins</groupId>
   <artifactId>maven-jar-plugin</artifactId>
   <version>3.3.0</version>
   <configuration>
    <archive>
     <manifest>
      <mainClass>App</mainClass>
     </manifest>
    </archive>
   </configuration>
  </plugin>

 </plugins>
</build>

</project>
```

4.Maven Build

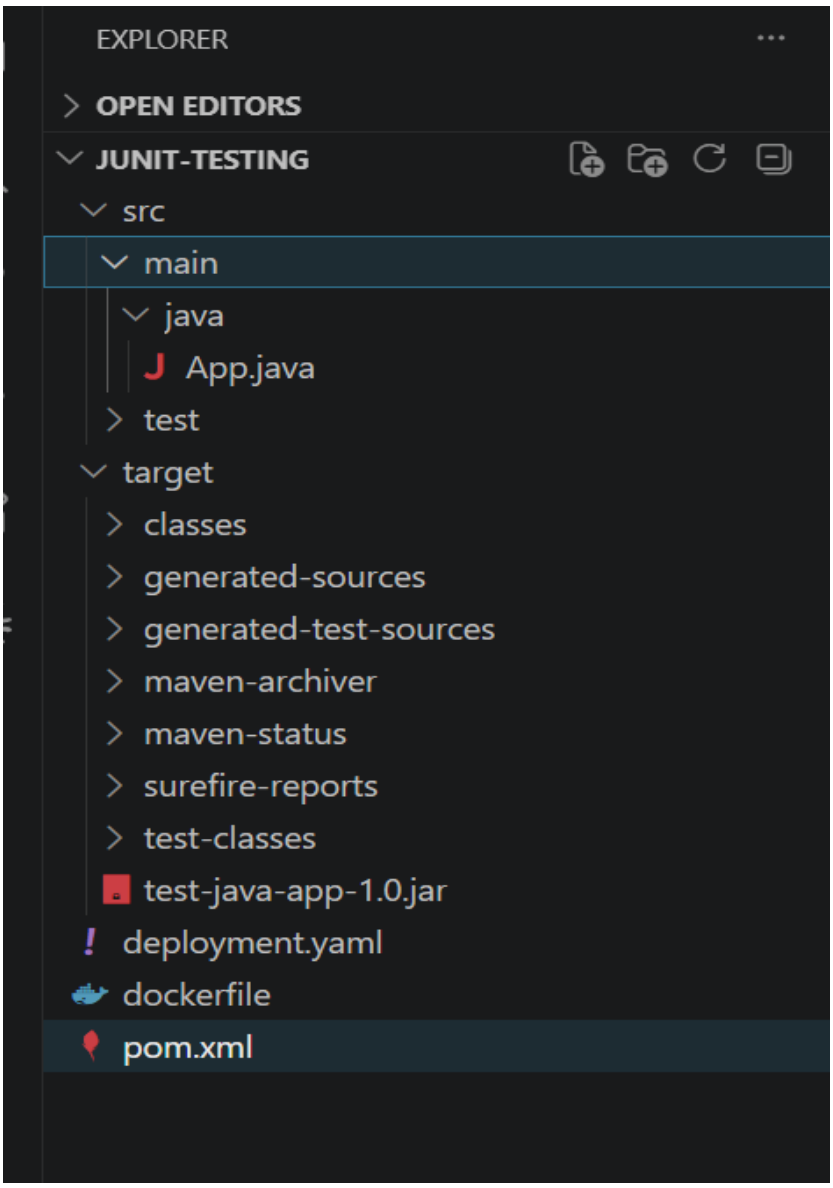
Command:

```
mvn clean test
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing
$ mvn clean test
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.956 s
[INFO] Finished at: 2026-04-15T18:01:00+05:30
[INFO] ------------------------------------------------------------------------

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing
$
```

mvn clean package

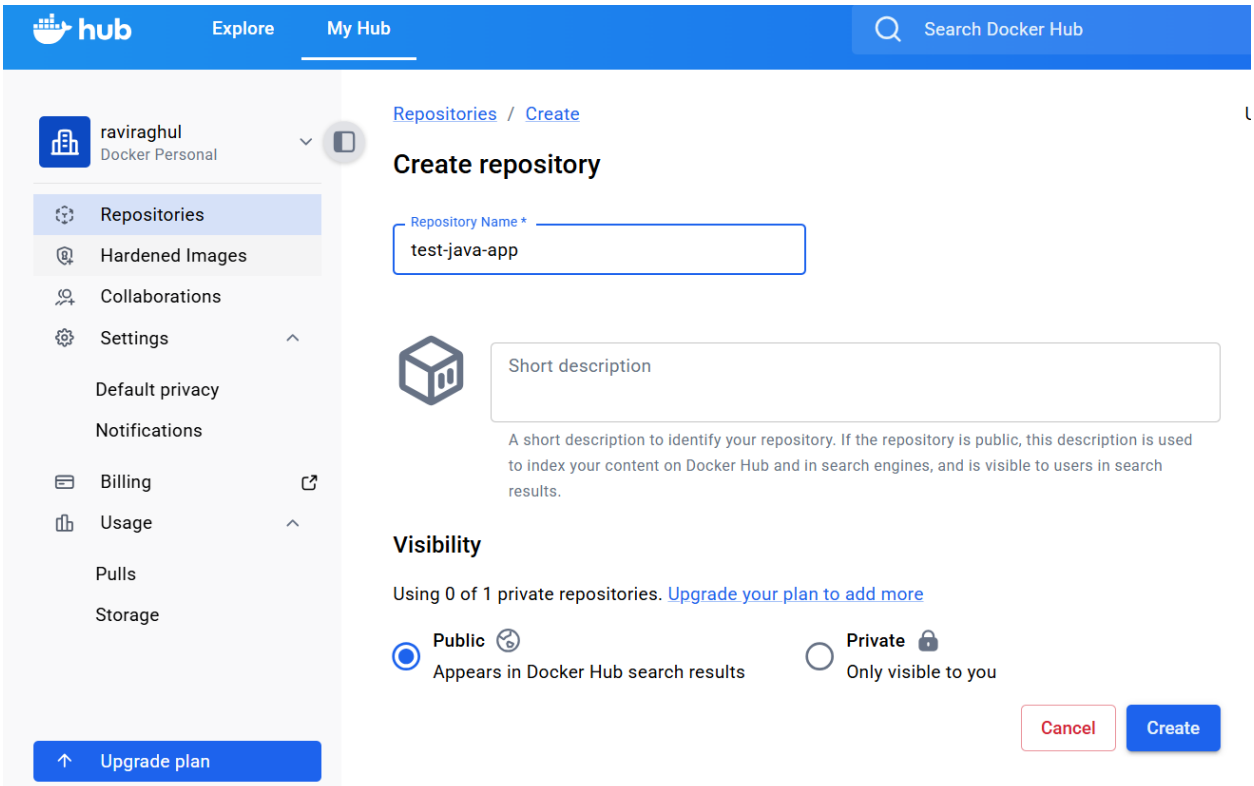
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing
$ mvn clean package
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\itsme\OneDrive\Desktop\Agile Lab Fat\JUNIT-Testing\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.745 s
[INFO] Finished at: 2026-04-15T18:01:16+05:30
[INFO] ------------------------------------------------------------------------

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing
$
```



5.Dockerfile

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

DockerHub Repository Created -




```
C:\Users\ravir>docker login
Authenticating with existing credentials... [Username: raviraghul]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded

C:\Users\ravir>
```

Commands -

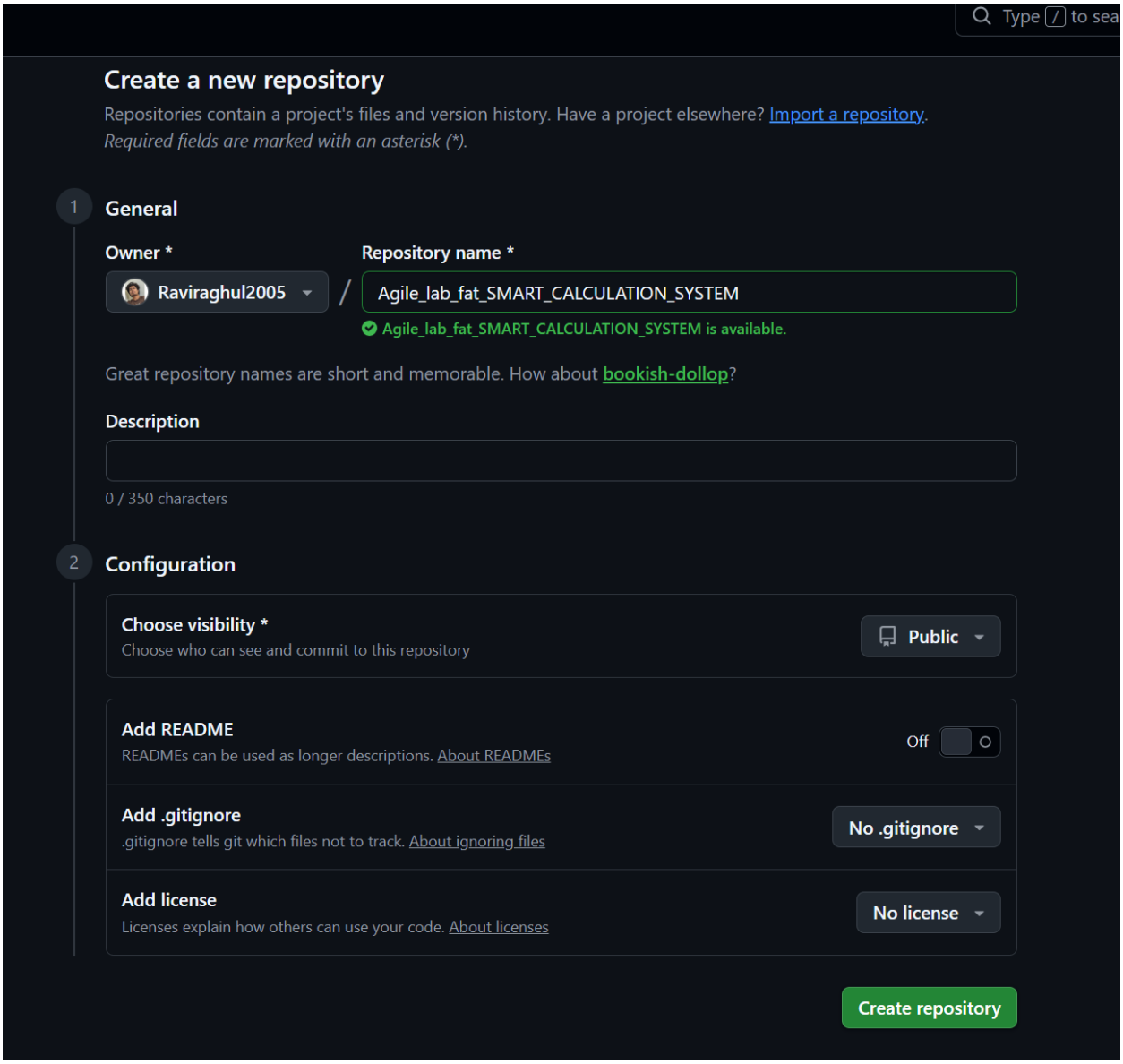
docker build -t raviraghul/test-java-app .

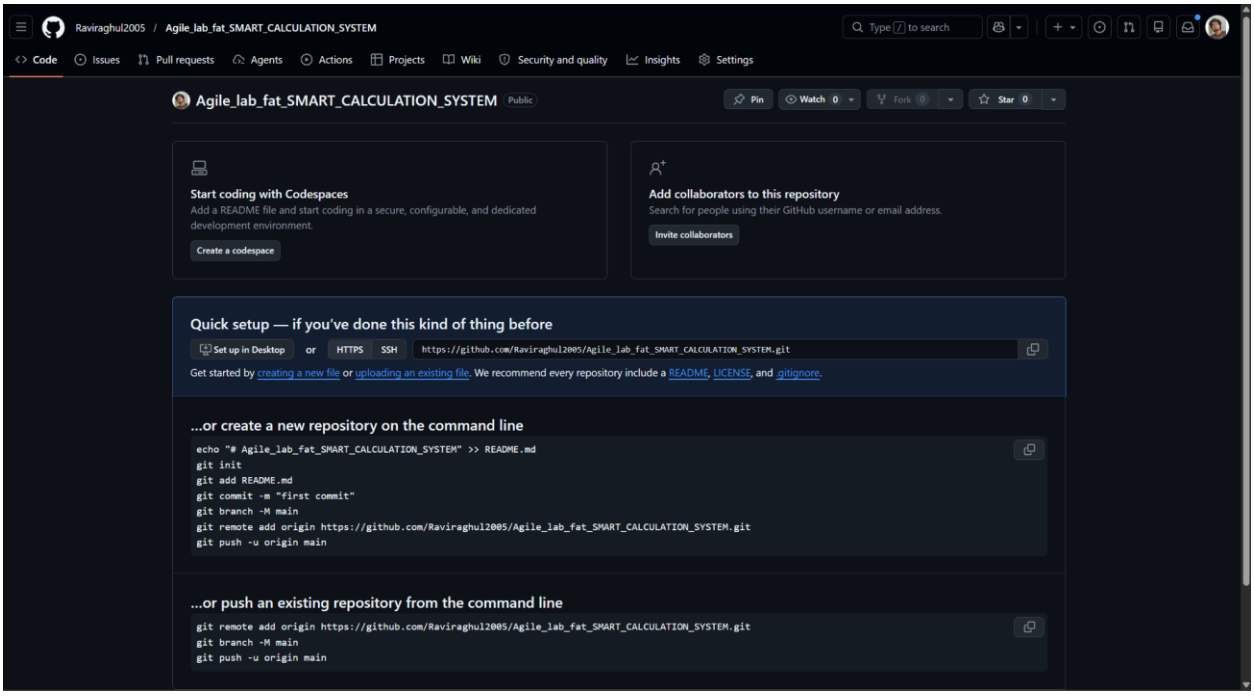
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ docker build -t raviraghul/test-java-app .
[+] Building 3.2s (9/9) FINISHED
 => [internal] load build definition from dockerfile
 => => transferring dockerfile: 167B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec4745125a47bd4f882
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec4745125a47bd4f882
 => [internal] load build context
 => => transferring context: 76B
 => CACHED [2/3] WORKDIR /app
 => CACHED [3/3] COPY target/*.jar app.jar
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:8f5f3db6c0a50edbfce9e447b9d7922318c7d8b4614c1b339682be06f56de58d
 => => exporting config sha256:b06005165ae0a2256b138d2d6be51e04485ca9ccc9c67e6a7ef8369e640740be
 => => exporting attestation manifest sha256:7f2f81603d258085e4c12f310fa24da563528565da23bba8be76368897fbeca1
 => => exporting manifest list sha256:c6d28c857479dc6ba2c95cfcc0aae86931e99f62a959c7c564c2c5cf75e302c0
 => => naming to docker.io/raviraghul/test-java-app:latest
 => => unpacking to docker.io/raviraghul/test-java-app:latest
```

docker push raviraghul/test-java-app

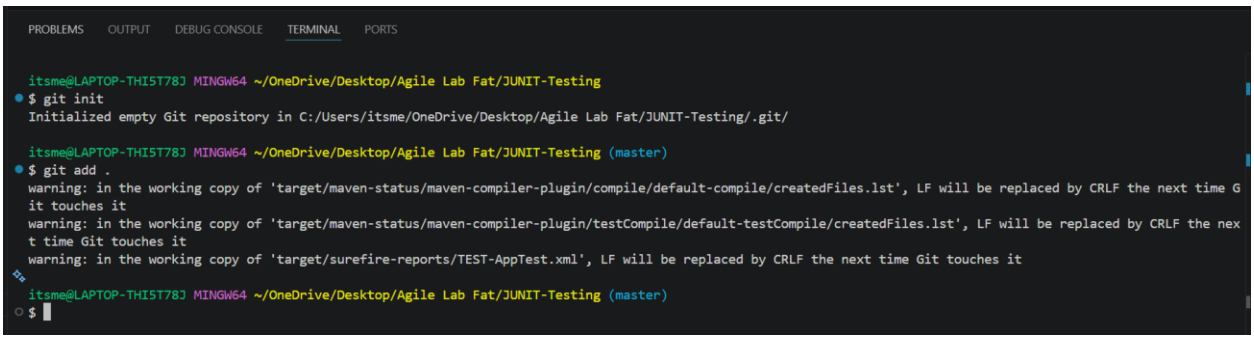
```
itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ docker push raviraghul/test-java-app
Using default tag: latest
The push refers to repository [docker.io/raviraghul/test-java-app]
946c51efe0fd: Layer already exists
a28b141e7dab: Layer already exists
82167420c5f0: Pushed
77ade34276c5: Layer already exists
3a547da4fe15: Layer already exists
82c540078c28: Layer already exists
7cd2436f004e: Layer already exists
589002ba0eae: Layer already exists
latest: digest: sha256:c6d28c857479dc6ba2c95cfcc0aae86931e99f62a959c7c564c2c5cf75e302c0 size: 856
```

6.Github Repo



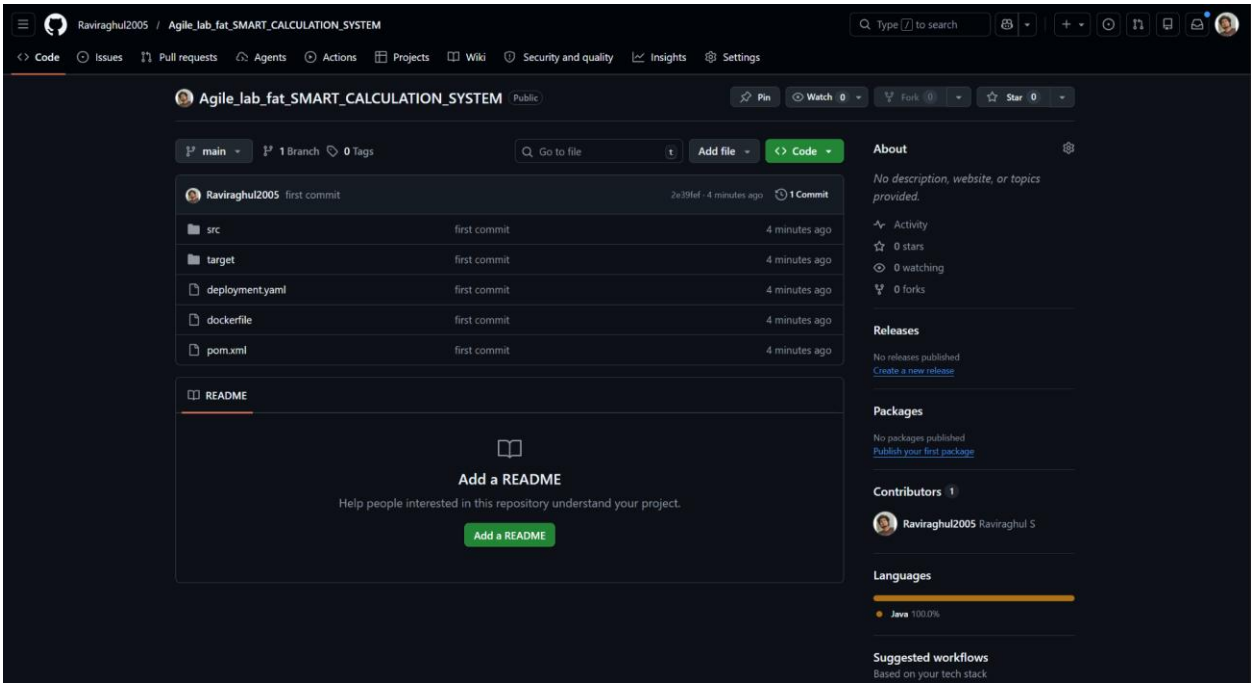


Git commands -



```
itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (master)
$ git commit -m "first commit"
[master (root-commit) 2e39fef] first commit
 15 files changed, 239 insertions(+)
 create mode 100644 deployment.yaml
 create mode 100644 dockerfile
 create mode 100644 pom.xml
 create mode 100644 src/main/java/App.java
 create mode 100644 src/test/java/AppTest.java
 create mode 100644 target/classes/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/AppTest.txt
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
 create mode 100644 target/test-java-app-1.0.jar

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (master)
$ git branch -M main

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ git remote add origin https://github.com/Raviraghul2005/Agile_lab_fat_SMART_CALCULATION_SYSTEM.git
```

```
itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ git push -u origin main
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 24 threads
Compressing objects: 100% (21/21), done.
Writing objects: 100% (33/33), 7.76 KiB | 567.00 KiB/s, done.
Total 33 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Raviraghul2005/Agile_lab_fat_SMART_CALCULATION_SYSTEM.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$
```


Raviraghul2005 / Agile_lab_fat_SMART_CALCULATION_SYSTEM
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Agile_lab_fat_SMART_CALCULATION_SYSTEM
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
Raviraghul2005 first commit
2e39fef · 4 minutes ago
1 Commit
src
first commit
4 minutes ago
target
first commit
4 minutes ago
deployment.yaml
first commit
4 minutes ago
dockerfile
first commit
4 minutes ago
pom.xml
first commit
4 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Raviraghul2005 Raviraghul S
Languages
Java 100.0%
Suggested workflows
Based on your tech stack

7.Kubernetes Deployment

deployment.yaml file

```yaml
apiVersion: apps/v1

kind: Deployment

metadata:

  name: java-test-app

spec:

  replicas: 2

  selector:

    matchLabels:

      app: java-test-app

  template:

    metadata:

      labels:

        app: java-test-app

    spec:

      containers:

      - name: java-test-app

        image: raviraghul/test-java-app:latest

        ports:

        - containerPort: 8080

---

apiVersion: v1
```

```yaml
kind: Service

metadata:

  name: java-test-service

spec:

  type: NodePort

  selector:

    app: java-test-app

  ports:

  - port: 80

    targetPort: 8080

    nodePort: 30007
```

Deploy to Kubernetes

`kubectl apply -f deployment.yaml`

```
itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ kubectl apply -f deployment.yaml
deployment.apps/java-test-app unchanged
service/java-test-service unchanged
```

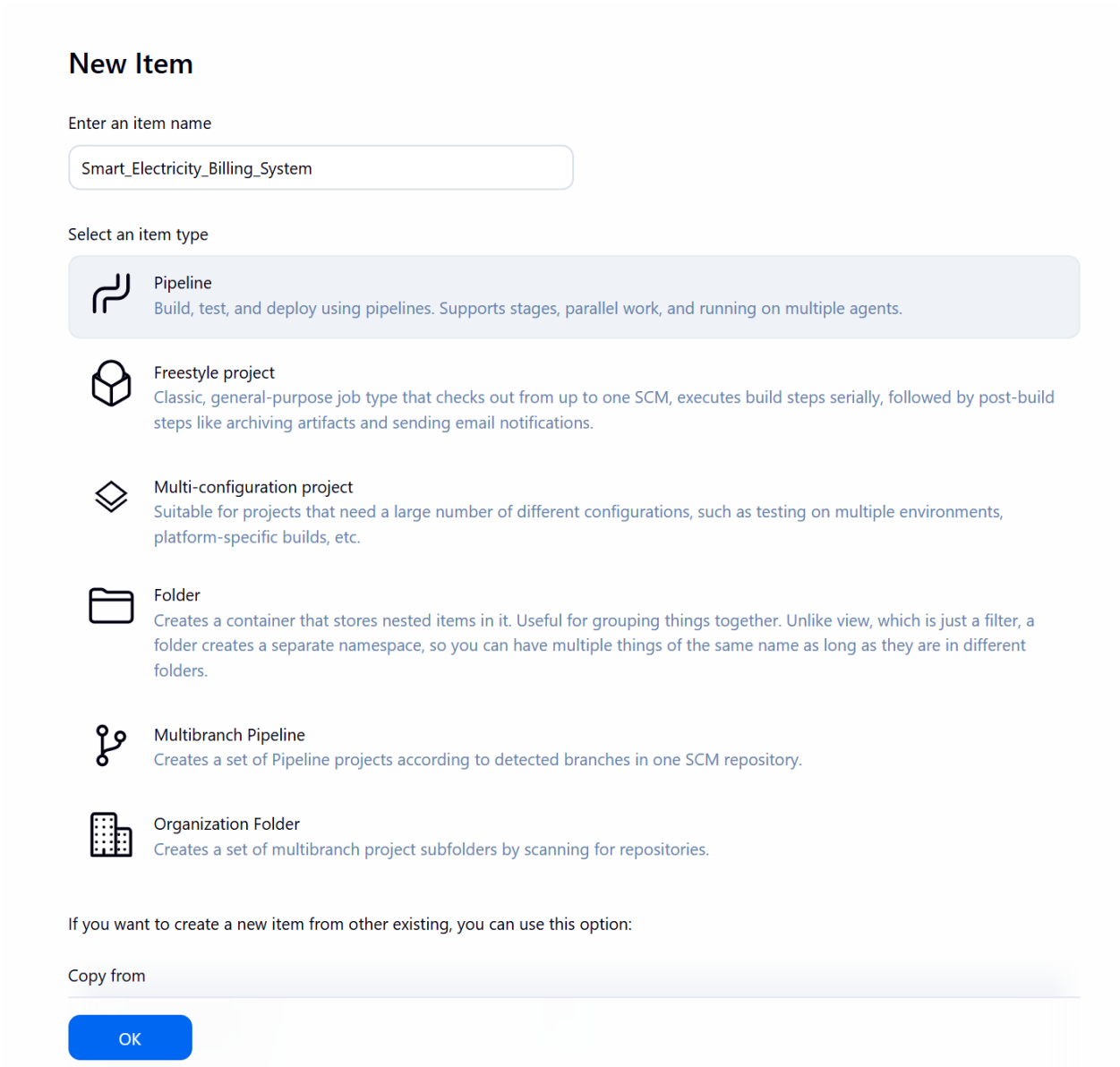
Verify Pods

`kubectl get pods`

```
itsme@LAPTOP-THI5T78J MINGW64 ~/OneDrive/Desktop/Agile Lab Fat/JUNIT-Testing (main)
$ kubectl get pods
NAME                              READY   STATUS             RESTARTS        AGE
java-test-app-7465c85cff-4ddvz    0/1     CrashLoopBackOff   74 (3m8s ago)   7d3h
java-test-app-7465c85cff-xwhkr    0/1     CrashLoopBackOff   74 (3m8s ago)   7d3h
```

Jenkins Pipeline

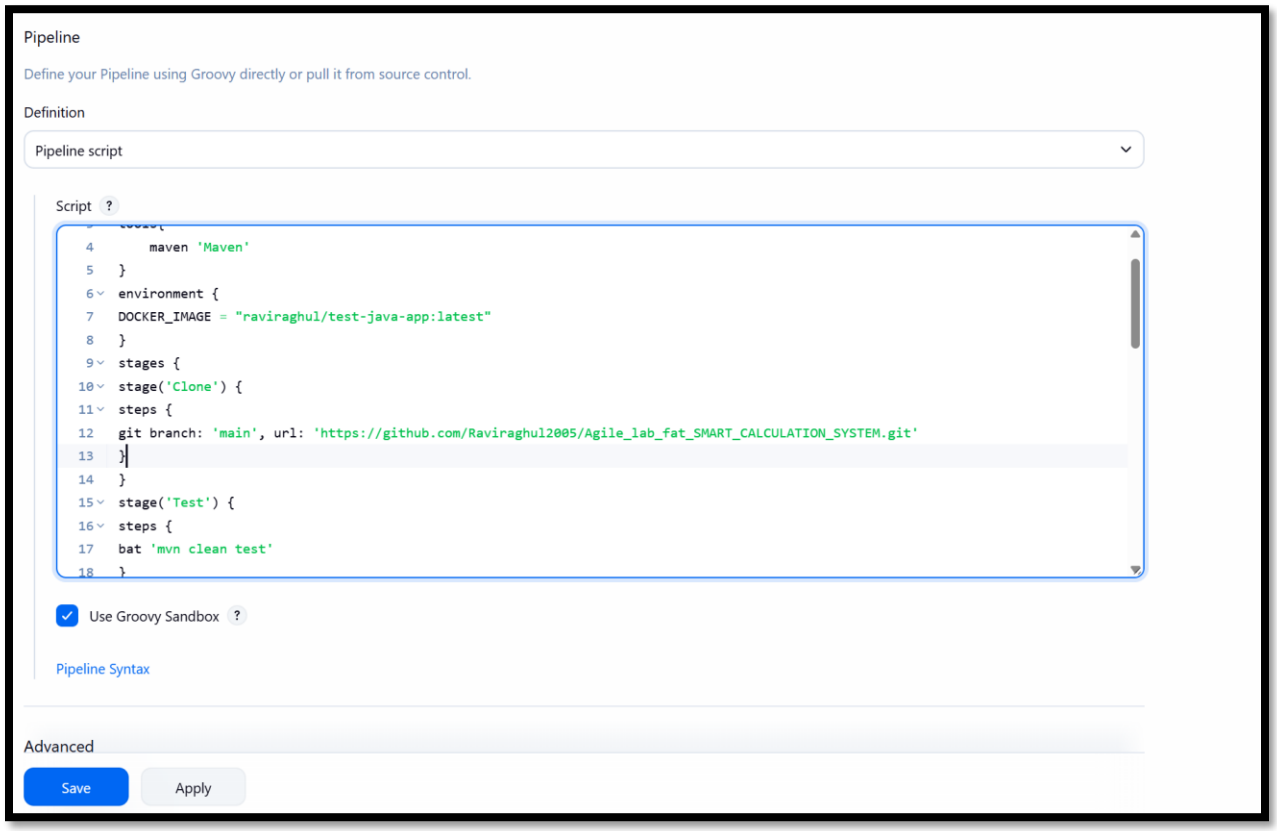
1.Create New Pipeline



2.Configure Pipeline

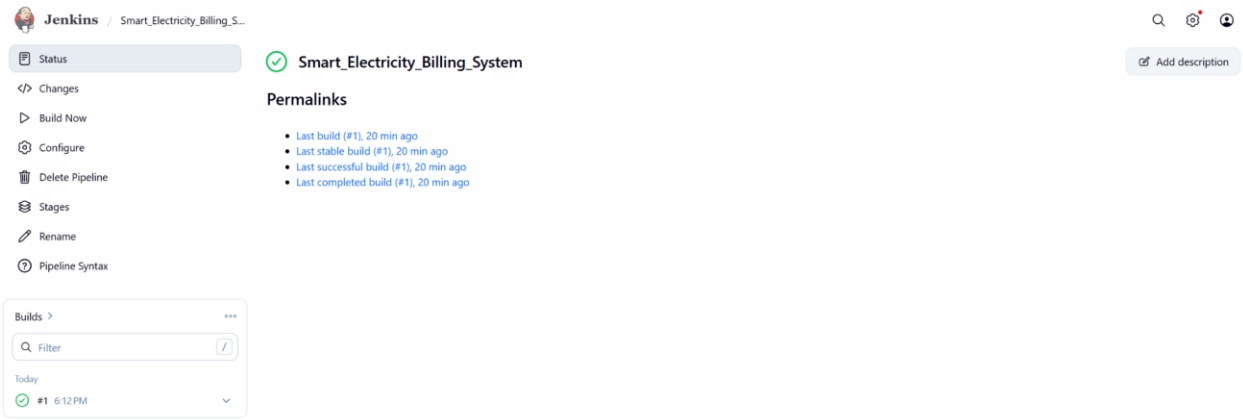
```
pipeline {
 agent any
 tools{
     maven 'Maven'
 }
 environment {
 DOCKER_IMAGE = "raviraghul/test-java-app:latest"
 }
 stages {
 stage('Clone') {
 steps {
 git branch: 'main', url:
'https://github.com/Raviraghul2005/Agile_lab_fat_SMART_CALCULATION_SYSTEM.git'
 }
 }
 stage('Test') {
 steps {
 bat 'mvn clean test'
 }
 }
 stage('Build') {
 steps {
 bat 'mvn package -DskipTests'
 }
 }
 stage('Docker Build') {
 steps {
 bat "docker build -t %DOCKER_IMAGE% ."
 }
 }
 stage('Push to DockerHub') {
 steps {
 withCredentials([usernamePassword(
 credentialsId: 'dockerhub-creds',
 usernameVariable: 'USER',
 passwordVariable: 'PASS'
 )]) {
 bat """
 docker login -u %USER% -p %PASS%
 docker push %DOCKER_IMAGE%
 """
 }
```

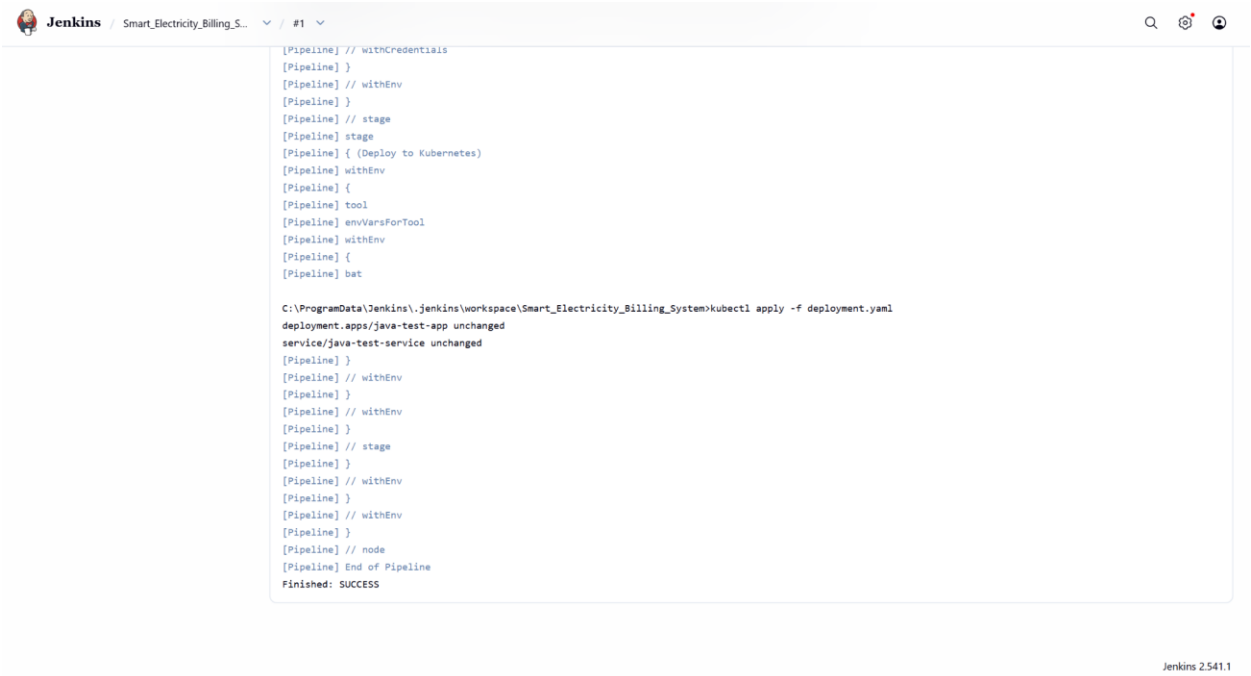
```
}
}
stage('Deploy to Kubernetes') {
    environment {
KUBECONFIG = "C:/Users/itsme/.kube/config"
}
steps {
bat 'kubectl apply -f deployment.yaml'
}
}
}
}
```



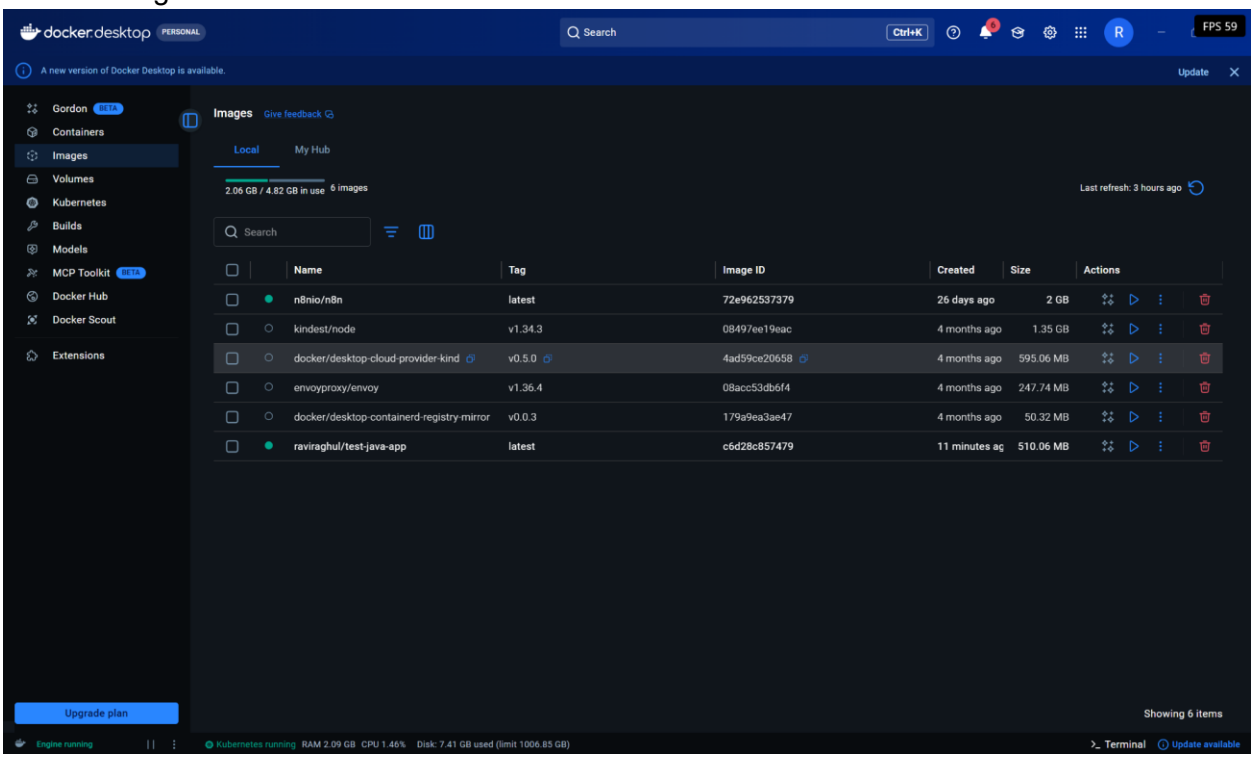
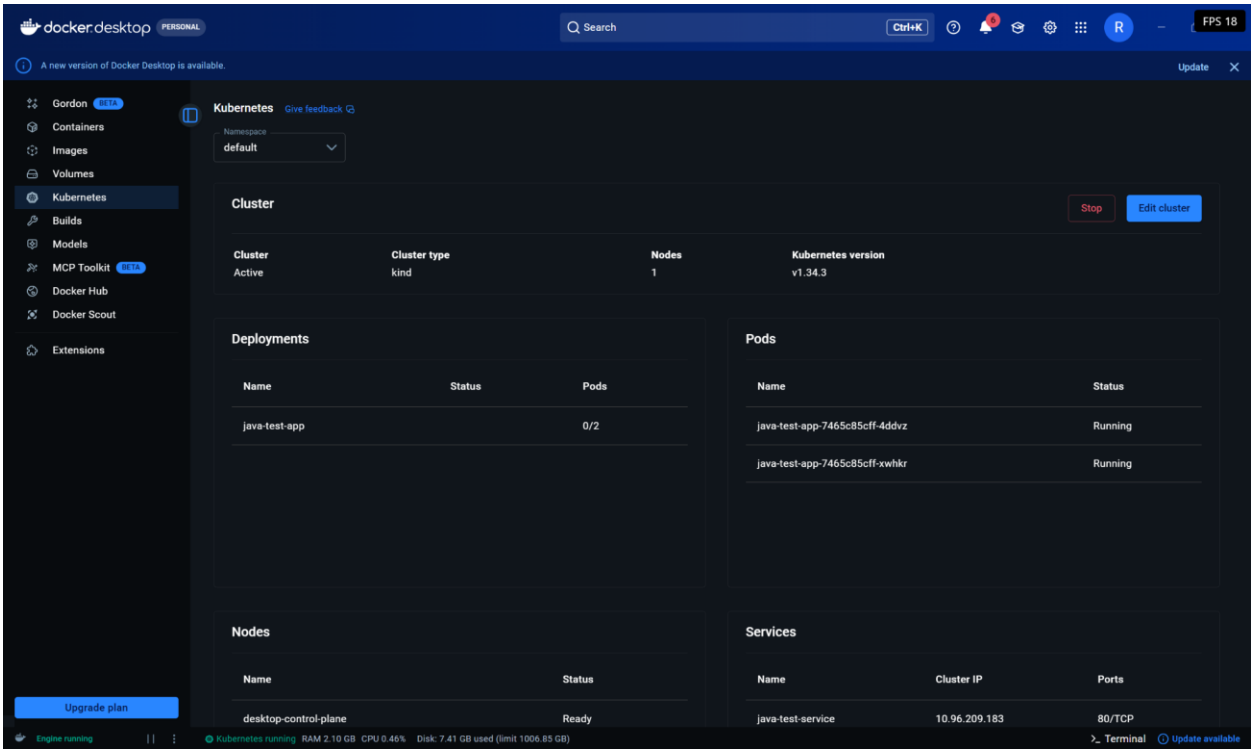
3.Save and Run

Final Jenkins Execution -





Docker Image



Kubernetes